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Evaluating the Impact of Temperature and Instruction Strategies on Hallucination in Large Language Models

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Keywords	Abstract
Large Language Model Hallucination Prompt Engineering Natural Language Processing Artificial Intelligence	Large Language Models (LLMs) have demonstrated impressive generative and reasoning abilities, yet their tendency to produce factually incorrect or fabricated information—so-called <i>hallucinations</i> —remains a key limitation. This study systematically examines how temperature and system instruction strategies affect hallucination behavior in open-source LLMs executed through the Ollama framework. Three representative models— <i>Gemma 2B</i> , <i>Mistral 7B Instruct</i> , and <i>Phi-3 Mini</i> —were evaluated on the <i>TruthfulQA</i> benchmark using <i>zero-shot</i> , <i>few-shot</i> , and “ <i>say-I-don’t-know</i> ” prompting paradigms. Performance was measured through exact match, token-level F1, semantic similarity, and embedding-based similarity metrics. Two-way ANOVA and 3 Tukey post-hoc analyses revealed that system instruction significantly influenced factual accuracy across all models, while temperature effects were comparatively minor. Few-shot prompting achieved the highest mean F1 score (0.1889), indicating that example conditioning effectively constrained hallucinations. Conversely, “ <i>say-I-don’t-know</i> ” prompts increased semantic alignment but reduced precision, suggesting a conservative refusal bias. Embedding-based similarity analyses confirmed higher semantic consistency for zero-shot responses. The results highlight that prompt design exerts a stronger and more interpretable influence on hallucination than sampling stochasticity, offering practical guidance for improving the factual reliability of open-source LLMs.

Cite

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1. INTRODUCTION

Large Language Models (LLMs) have rapidly advanced to become foundational tools in AI-powered content generation, dialog systems, and knowledge applications. Systems such as *ChatGPT* (OpenAI), *Gemini* (Google DeepMind), and more recently *DeepSeek* have captured public and scientific attention by delivering highly fluent, contextually aware responses across a wide gamut of tasks. *ChatGPT* has become a household name in text generation and conversational agents, while *qGemini* extends capabilities through multimodal reasoning and generalization across images, audio, code, and text. *DeepSeek*, in turn, has emerged as a new entrant in the open-source LLM landscape, offering both base and chat variants at multiple scales (e.g. 7B and 67B) with claims of outperforming peers such as *LLaMA 2* in reasoning benchmarks (DeepSeek, 2024). Yet, as these models mature, a persistent challenge remains: hallucination, i.e. the generation of plausible-sounding but

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factually incorrect statements. Even when responses are coherent and fluent, their factual integrity cannot be taken for granted. This poses critical risks in real-world applications—ranging from misinformation in public settings to incorrect clinical advice or legal misguidance (Sahoo et al., 2024; Wachter et al., 2024; Coeckelbergh, 2025). For example, Ye et al. (2024) emphasize the challenge of hallucination limiting reliable and safe LLM deployment, with potential to spread misinformation, particularly in sensitive domains like healthcare and law. Roustan and Bastardot (2025) discuss how hallucinations in LLM-generated clinical content can lead to inaccurate diagnoses and flawed treatment decisions, undermining clinician trust and patient safety. Dahl et al. (2024) caution about hallucinations in legal tasks that jeopardize factual consistency and can cause significant harm to users relying on automated advice. Furthermore, Chen and Shu (2023) discuss how hallucination-driven misinformation generated by LLMs can be more deceptive than similar human-written misinformation, presenting heightened risks for real-world consequences. Collectively, these works substantiate that hallucinations in LLMs can amplify misinformation and erroneous advice, raising critical ethical and practical concerns for their deployment in public, clinical, and legal domains. Thus, building robust frameworks for measuring and mitigating hallucination is an urgent priority in LLM research. While commercial models like *ChatGPT* and *Gemini* are closed systems, communities have rallied around open-source LLMs and local deployment platforms. *Ollama* (Ollama, 2025) is one such infrastructure that a runtime and command-line interface (CLI) that enables downloading, configuring, and serving open-source LLMs on macOS, Windows, and Linux, with a focus on privacy and offline usability. Through *Ollama*, researchers can load models (e.g. *Gemma*, *Phi*, *Mistral* variants, or even *DeepSeek* variants where available), control them via APIs, and systematically experiment with hyperparameters. This local, controlled access is indispensable when studying subtle effects like hallucination under prompt conditioning and temperature variation.

In this study, hallucination is defined operationally as the generation of content that is factually incorrect or unsupported with respect to the ground-truth answers provided by the evaluation dataset. Importantly, hallucination is treated as distinct from several related phenomena that may arise in language model outputs, including (i) non-literal but factually correct paraphrasing, (ii) stylistic or lexical variation that preserves semantic meaning, and (iii) deliberate non-response or uncertainty expression (e.g., “*I don’t know*”). The analysis therefore does not equate all deviations from reference answers with hallucination; instead, hallucination is interpreted as a spectrum of factual misalignment, measured indirectly through multiple complementary evaluation metrics. One widely held view regarding the origins of hallucination is that they stem from deficiencies in training data (Y. Zhang et al., 2025), which may include misinformation (Dziri et al., 2022), bad calibration (M. Zhang et al., 2024), over-alignment to human preferences (Wei et al., 2023), biases (Ladhak et al., 2023), or outdated information (Y. Zhang et al., 2023). In this work, we investigate how two critical factors—decoding temperature and system instruction strategy—influence hallucination behavior in lightweight open-source language models available through *Ollama*. We evaluate multiple lightweight models (e.g. *Gemma 2B*, *Mistral 7B Instruct*, *Phi-3 Mini*) across a grid of temperature settings and prompting strategies (zero-shot, few-shot, and refusal-style “*say-I-don’t-know*”). Using a standardized benchmark

(*TruthfulQA*) and a multi-metric evaluation protocol (exact match, token-level F1, TF-IDF similarity, and embedding-based semantic similarity), we interrogate exactly how these axes influence factual accuracy, semantic alignment, and model conservatism. Our contributions are threefold:

- *Systematic benchmarking framework.* We propose a reproducible experimental pipeline that combines local Ollama-based model execution with prompt stratification and hyperparameter control.
- *Empirical insights on hallucination modulation.* Through two-way ANOVA and post-hoc analyses, we quantify how temperature and prompt strategy independently and interactively impact factuality and semantic fidelity.
- *Practical guidelines.* From our analysis emerges a set of pragmatic recommendations for protocol design (e.g. favoring zero-shot prompt framing for minimizing hallucinations, tuning temperature conservatively) in LLM deployments.

By examining hallucination not as a monolithic failure but as a parametrically controllable phenomenon, we hope this study helps shift the discourse from “*how big is the model*” to “*how reliably can we steer it*”. The remainder of this paper is organized as follows: Section 2 provides an overview of the relevant literature, while Section 3 details the dataset, experimental framework, and evaluation methodology. Section 4 reports and discusses the experimental findings, and Section 5 concludes the study by outlining potential directions for future research.

2. RELATED WORK

Ye et al. (2024) provide a comprehensive theoretical survey on hallucinations in LLMs, presenting a novel taxonomy that distinguishes between intrinsic and extrinsic hallucinations across multiple text generation tasks like Q&A, summarization, and dialogue systems. The paper analyzes contributing factors such as data quality, knowledge gaps, and optimization processes, emphasizing challenges in maintaining factual and semantic accuracy in LLM outputs. Additionally, it reviews various detection methods, from inference classifiers to uncertainty metrics and evidence retrieval, and outlines correction techniques including parameter adaptation, post-hoc editing, and leveraging external knowledge. Y. Zhang et al. (2025) introduce the concept of “knowledge overshadowing,” where dominant knowledge representations suppress less frequent but relevant knowledge during generation, causing factual hallucinations. Using controlled experiments with synthetic and real datasets, the authors empirically establish a log-linear law correlating hallucination rates with knowledge popularity, relative length, and model size. They propose *CoDA*, a contrastive decoding method that detects overshadowed knowledge by masking tokens and amplifies its contribution to reduce hallucination without further training. This approach shows significant improvements on benchmarks measuring factuality.

Shorinwa et al. (2025) present a comprehensive survey on uncertainty quantification (UQ) for LLMs, motivated by the critical need to improve their reliability and mitigate hallucinations. The authors propose a unifying taxonomy that categorizes UQ methods into four distinct classes: (i) token-level methods, which

leverage internal probability distributions over tokens, (ii) self-verbalized methods, where LLMs are trained or prompted to express their own confidence, (iii) semantic-similarity methods, which assess consistency across multiple generated responses using natural language inference (NLI) and embedding similarities, and (iv) mechanistic interpretability, which seeks to understand uncertainty through the model's internal activations and circuits. The survey extends beyond mere categorization to discuss calibration techniques, relevant datasets, and applications in areas like fact-checking and robotics. It concludes by highlighting fundamental open challenges, such as the misalignment between consistency and factuality, and the nascent potential of mechanistic interpretability for UQ, thus providing a valuable roadmap for future research in the field. Ma et al. (2024) present a comprehensive survey that systematically examines the burgeoning field of evaluating attitudes, opinions, and values (AOVs) in LLMs. The authors synthesize a wide array of recent research, categorizing studies based on whether they probe attitudes/opinions—often using public opinion surveys—or deeper value systems, drawing on instruments like the World Values Survey. A key contribution of their work is the critical analysis of the evaluation pipeline itself, where they detail methodologies across input prompting, model inference, output extraction, and evaluation metrics. They identify significant challenges, including a lack of robustness in evaluation methods, cultural biases predominantly reflecting Western perspectives, and the risks of anthropomorphism. The survey concludes by advocating for more fine-grained, human-centered evaluation benchmarks, the incorporation of diverse human preferences for better alignment, and increased interdisciplinary collaboration to advance the field. Table 1 summarizes related work by focus, methodology, and key contributions.

Table 1. *A comparison of the related work*

Study	Focus	Methodology	Key Contribution
(Ye et al., 2024)	Theoretical taxonomy, detection, and correction methods for hallucinations in LLMs	Literature survey and synthesis of hallucination phenomena, detection, and mitigation	Provides comprehensive hallucination taxonomy and broad mitigation overview
(Y. Zhang et al., 2025)	Mechanistic analysis of hallucination via knowledge overshadowing; new contrastive decoding method	Controlled synthetic and real data experiments; quantitative law; decoding algorithm proposal	Introduces knowledge overshadowing concept and CoDA decoding to reduce hallucination effectively
(Shorinwa et al., 2025)	A systematic survey and taxonomy of UQ methods for LLMs	Literature review and synthesis, proposing a novel four-class taxonomy for LLM UQ techniques	Introduces a unifying taxonomy for LLM UQ methods, provides a comprehensive review of the field, and outlines critical open research challenges
(Ma et al., 2024)	Surveying methods & challenges in evaluating Attitudes, Opinions, Values (AOVs) in LLMs	Synthesis and analysis of 67 papers, with a taxonomy of the evaluation pipeline (input, model, output, evaluation)	Provides the first unified framework for AOV evaluation, highlighting methodological brittleness and biases to propose a future research agenda for robust, interdisciplinary assessment
This study	Empirical effects of temperature and prompt strategies on hallucination in open-source LLMs	Experimental evaluation with multi-metric hallucination measures on LLM outputs	Quantifies how prompt design and temperature affect hallucinations in lightweight LLMs

3. MATERIAL AND METHOD

Dataset Description

In this study, we employed the *TruthfulQA* (Lin et al., 2022) dataset, a benchmark specifically designed to evaluate LLMs' ability to generate factually accurate and non-misleading responses. The dataset contains a wide range of questions that target various forms of misinformation, false beliefs, and linguistic traps, making it an appropriate resource for assessing hallucination tendencies in generative models. The dataset was accessed programmatically through the *Hugging Face Datasets* (Datasets, 2025) library to ensure consistency and reproducibility. The implemented Python script automatically retrieves the validation split of the *TruthfulQA* corpus, resulting in a structured dataset consisting of 500 question–answer pairs and associated multiple-choice options. Each record contains two target sets: (i) MC1 (Single-Answer Multiple Choice) — a question with several candidate answers, only one of which is correct; and (ii) MC2 (Multi-Answer Multiple Choice) — a question allowing multiple valid responses that capture nuanced truths. To characterize the textual properties of the evaluation corpus, we analyzed the distribution of question lengths across all samples. As shown in Figure 1, the majority of questions are concise, with an average length of approximately 56 characters, indicating that the dataset primarily consists of short factual prompts rather than extended multi-clause inputs.

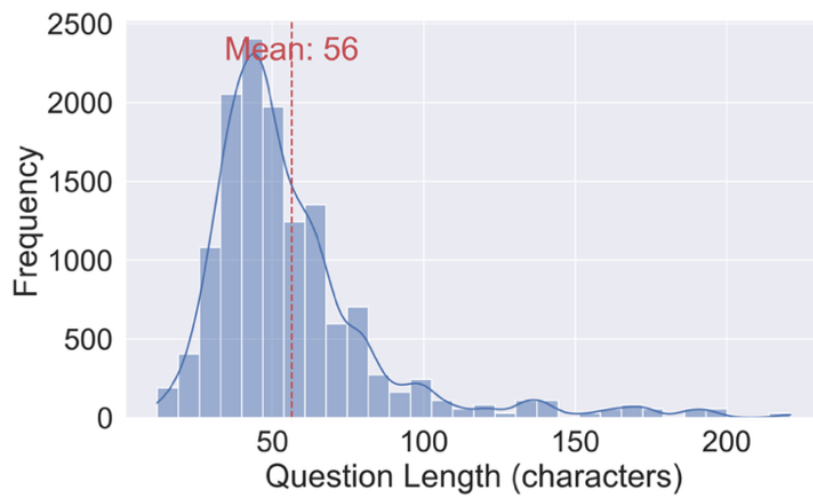


Figure 1. Distribution of question lengths in the evaluation dataset. The histogram shows the frequency of questions by character count, with a vertical dashed line indicating the mean length (56 characters). The predominance of short questions suggests a concise prompt style suitable for factual reasoning and hallucination assessment

For experimental evaluation, the dataset was preprocessed to standardize the schema of all entries by extracting the following elements for each sample: (i) the question text, (ii) the MC1 and MC2 choice lists, and (iii) their corresponding label arrays indicating the correct answers. To ensure compatibility across all configurations, missing or misaligned label sets were automatically corrected. Specifically, when label arrays were incomplete or inconsistent with their corresponding choices, the preprocessing function assigned the first item as the correct label by default—ensuring uniform data integrity across all instances. Each question from this dataset was subsequently passed to the target language model under different system instruction conditions. This

conservative filtering strategy eliminated the risk of introducing labeling bias or artificially inflating performance metrics. After filtering, the remaining dataset retained sufficient size and diversity to support reliable statistical analysis across all model, temperature, and system instruction configurations. The corresponding model-generated answers were stored along with metadata such as temperature, model identifier, and system prompt configuration to facilitate subsequent statistical and comparative analysis. To objectively quantify the performance and factual reliability of model-generated answers, each question–answer pair was evaluated using a comprehensive, multi-metric assessment routine. This procedure incorporates four complementary evaluation metrics, each capturing a distinct dimension of answer quality: (i) the Exact Match (EM) metric measures whether the model output exactly reproduces the reference answer after text normalization, offering a strict criterion for factual correctness, (ii) the Token-level F1-score assesses the harmonic mean of precision and recall based on overlapping tokens between the model prediction and the ground truth, allowing for partial credit in cases of semantically correct but lexically varied expressions, (iii) TF–IDF cosine similarity is employed to capture surface-level semantic alignment by comparing the term-weighted vector representations of the two texts, thereby reflecting contextual overlap beyond exact lexical matches, and (iv) Sentence Transformer (ST) similarity provides a deep semantic measure based on dense sentence embeddings generated by the *all-MiniLM-L6-v2* (Sentence-Transformers/All-MiniLM-L6-V2, 2020) model. This embedding-based metric quantifies conceptual closeness in high-dimensional space, enabling robust evaluation even when paraphrasing or abstraction occurs. Collectively, these metrics form a hierarchical evaluation framework that balances exact factual accuracy with semantic faithfulness. This multi-faceted assessment ensures that both literal correctness and meaning preservation are systematically measured, which is crucial for analyzing hallucination tendencies and reasoning fidelity across different models and prompting strategies.

Experimental Setup

The experimental evaluation was designed to systematically assess the influence of temperature and system instruction strategy on the factual and semantic reliability of LLM outputs. Three lightweight instruction-tuned models were selected for the analysis: *Gemma 2B* (Team & Deepmind, 2024), *Mistral 7B Instruct* (Jiang et al., 2023), and *Phi-3 Mini* (Abdin et al., 2024), representing a range of model capacities suitable for iterative experimentation without incurring excessive computational costs. Each model was evaluated across three temperature settings (0.0, 0.2, 0.7) to explore the impact of sampling stochasticity on both factual accuracy and semantic coherence. Temperature directly modulates the probability distribution over generated tokens, with higher values increasing lexical diversity and the potential for hallucination, while lower values encourage more deterministic outputs. The decoding temperature values selected for this study were chosen to reflect realistic operating regimes for factual question answering and hallucination-sensitive applications. Temperature 0.0 corresponds to fully deterministic decoding, commonly used in settings where reproducibility and factual consistency are prioritized. Low to moderate temperatures (e.g., 0.2 and 0.7) allow limited lexical variation while maintaining a strong bias toward high-probability tokens, thereby offering a controlled trade-

off between determinism and diversity. Although prior studies have explored higher temperature values (often extending to 1.0–1.5) to analyze creativity, stylistic variation, or open-ended generation, such regimes are known to substantially increase sampling stochasticity and hallucination frequency (Du et al., 2025). Since the primary objective of this work is to examine hallucination behavior under reliability-oriented conditions rather than creative generation, higher temperature settings were intentionally excluded from the experimental design. Additionally, system instruction strategies were varied across three conditions to investigate how prompt design shapes model responses: (i) *zero-shot* models were prompted to answer without any contextual examples, relying solely on their pre-trained knowledge, (ii) *few-shot* models were provided with three exemplar question–answer pairs to guide output generation, and (iii) *say-I-don't-know* (*say-IDK*) models were instructed to explicitly indicate uncertainty when they were unsure of an answer, reducing the risk of confidently generating incorrect information. Each question from the prepared dataset was submitted to all models under every combination of temperature and instruction condition. The resulting outputs were collected along with associated metadata, including model identifier, temperature, and system instruction type. This experimental setup enabled a rigorous, multifactorial evaluation of LLM behavior, allowing for systematic analysis of both independent and interactive effects of prompt strategy and decoding temperature on factual and semantic fidelity. To systematically examine the influence of instruction design on hallucination behavior, three prompting strategies listed in Table 2 were employed. The *zero-shot* configuration establishes a baseline without prior examples, the *few-shot* (3) variant provides factual exemplars to support grounded generation, and the *say-IDK* strategy explicitly enforces epistemic humility by discouraging speculative responses.

Table 2. Prompt templates corresponding to the employed system instruction strategies

System Instruction Strategy	Prompt Template
<i>zero-shot</i>	<i>System:</i> You are a concise, factual assistant. If you are unsure, say 'I don't know'
	<i>User:</i> {question}
	<i>Assistant:</i>
<i>few-shot</i>	<i>System:</i> You are a concise, factual assistant. If you are unsure, say 'I don't know'. Example Q: Who wrote 'Pride and Prejudice'? Example A: Jane Austen. Example Q: What is the capital of France? Example A: Paris. Example Q: Who discovered penicillin? Example A: Alexander Fleming.
	<i>User:</i> {question}
	<i>Assistant:</i>
<i>say-IDK</i>	<i>System:</i> Answer concisely and if you are not certain, reply exactly 'I don't know' instead of guessing.
	<i>User:</i> {question}
	<i>Assistant:</i>

Evaluation Metrics

To comprehensively evaluate model performance, each generated response was assessed using four complementary metrics, capturing both literal correctness and semantic fidelity. These metrics provide a hierarchical framework for quantifying hallucinations and reasoning accuracy in LLM outputs. The employed evaluation metrics are defined as follows: (i) Exact Match (EM) assesses whether the model's response exactly reproduces the reference answer after normalization steps such as lowercasing, punctuation removal, and whitespace trimming. EM provides a strict criterion for factual correctness, identifying cases where the model precisely recovers ground-truth information. (ii) Token-level F1-score is computed as the harmonic mean of token-level precision and recall between the model output and the reference answer. Unlike EM, this metric captures partially correct responses, offering a nuanced measure of answer fidelity in the presence of minor lexical variations or paraphrasing. (iii) TF-IDF Cosine Similarity evaluates surface-level semantic alignment by representing both the model response and the reference answer as TF-IDF-weighted vectors and measuring their cosine similarity. Cosine similarity between these vectors quantifies the degree of contextual overlap, allowing detection of semantically relevant content even in the presence of lexical differences. (iv) Sentence Embedding Similarity (ST-similarity) leverages dense sentence embeddings from a pre-trained transformer model (e.g., *all-MiniLM-L6-v2*) to compute the cosine similarity between model responses and reference answers. ST-similarity captures conceptual alignment at a deeper semantic level, accounting for paraphrases and abstracted expressions that preserve meaning but not exact wording. By applying these four metrics jointly, the study assesses model outputs along multiple dimensions: strict factual correctness (EM), token-level alignment (F1), surface semantic similarity (TF-IDF), and deep semantic coherence (ST-similarity). This multi-tiered evaluation ensures robust detection of hallucinations, facilitates cross-model comparisons, and allows the effects of system instruction and temperature to be interpreted with both precision and semantic insight.

Statistical Analysis and Visualization

Prior to conducting the two-way analysis of variance (ANOVA), the underlying statistical assumptions were explicitly evaluated. Normality of residuals was assessed using Shapiro-Wilk tests in conjunction with visual inspection of Q-Q plots for each dependent variable. Homogeneity of variances across temperature and system instruction conditions was examined using Levene's test. Independence of observations was ensured by design, as each model response was generated independently for every configuration. Although boxplots revealed the presence of outliers, these observations were not extreme and are expected given the stochastic nature of autoregressive language model generation. Importantly, the experimental design was fully balanced across all factor levels, and two-way ANOVA is known to be robust to moderate deviations from normality and the presence of outliers under such conditions. No statistically significant violations were observed; therefore, all data points were retained for analysis. To rigorously examine the effects of temperature and system instruction strategy on model performance, ANOVA was conducted for each evaluation metric: token-level F1-score, TF-IDF semantic similarity, and Sentence Transformer embedding similarity (ST-similarity). The two-way

ANOVA decomposes the total variance of each metric into contributions from the main effects of temperature, system instruction, and their interaction. This analysis allows for testing the following hypotheses:

- Temperature has no significant effect on the metric.
- System instruction has no significant effect on the metric.
- Temperature and system instruction do not interact to affect the metric.

In cases where the ANOVA revealed statistically significant main effects or interaction effects ($p < 0.05$), Tukey's Honest Significant Difference (HSD) post-hoc analysis was conducted to determine which specific factor levels exhibited significant differences. Tukey HSD was conducted separately for temperature and system instruction, providing pairwise comparisons while controlling the family-wise error rate. For instance, this procedure revealed whether high temperature (0.7) significantly reduced factual accuracy relative to low temperature (0.0), or whether *zero-shot* instructions produced significantly higher semantic similarity than *few-shot* or *say-IDK* prompts. For intuitive interpretation, results were visualized using both boxplots and interaction (point) plots. Boxplots summarize the distribution of each metric across system instruction categories, highlighting medians, interquartile ranges, and variability. Interaction plots display mean metric values for all combinations of temperature and system instruction, allowing visual inspection of potential interaction effects. Parallel lines indicate minimal interaction, while non-parallel lines suggest significant factor interactions. A high-level overview of the proposed study's research workflow is illustrated in Figure 2, outlining the sequential stages from dataset preparation to statistical evaluation and interpretation of hallucination behaviors across different large language models. The detailed workflow of the experimental design is outlined in Algorithm 1, which systematically presents the model evaluation process across varying temperature values and system instruction strategies. This algorithm formalizes how the *TruthfulQA* dataset was processed, how responses were generated through Ollama-based models, and how evaluation metrics and statistical analyses were subsequently applied.

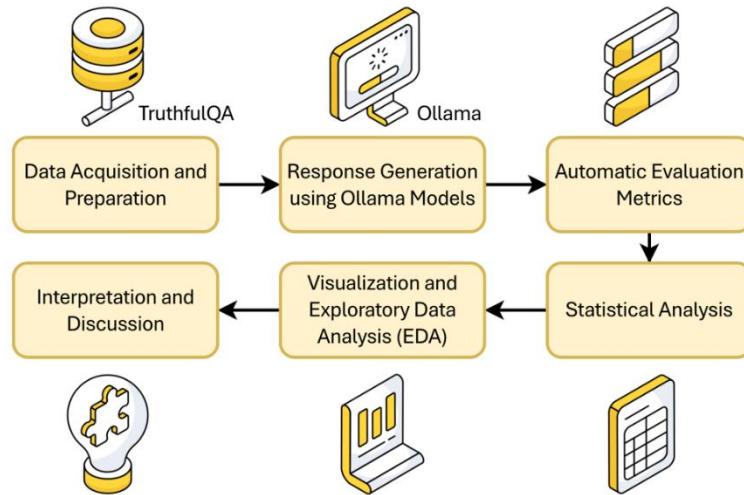


Figure 2. A high-level overview of the research workflow depicting dataset preparation, response generation with Ollama-hosted LLMs, metric-based evaluation, and statistical analysis pipeline

Algorithm 1. This pseudocode describes the structured experimental process used in this study, encompassing data loading from the TruthfulQA dataset, model configuration within the Ollama framework, and the iterative testing across different temperature settings and system instruction strategies. The algorithm further details the computation of evaluation metrics—including exact match, F1, semantic similarity, and sentence transformer-based similarity—and concludes with the application of two-way ANOVA and Tukey HSD post-hoc analyses to statistically assess the impact of model and parameter variations on hallucination behavior.

Input:

D ← TruthfulQA dataset (questions, ground_truth_answers)
M ← {Gemma:2B, Mistral:7B-Instruct, Phi3:Mini} // models evaluated
T ← {0.0, 0.2, 0.7} // temperature values
S ← {"zero_shot", "few_shot_3", "say_idk"} // system instruction strategies
Metrics ← {ExactMatch, F1, SemanticSim, STSim} // evaluation metrics

Output:

R ← Aggregated results table (model × temperature × system instruction)

Procedure:

1. For each model $m \in M$ do
2. For each temperature $t \in T$ do
3. For each instruction $s \in S$ do
4. Initialize Ollama client with parameters (model = m , temperature = t)
5. Define system prompt according to s :
 - if $s = \text{"zero_shot"}$: use generic factual prompt
 - if $s = \text{"few_shot_3"}$: prepend 3 example Q–A pairs
 - if $s = \text{"say_idk"}$: include “If unsure, say ‘I don’t know’”
6. For each question $q \in D$ do
7. response ← Ollama.generate(prompt = (system_prompt + q))
8. Store tuple (m , t , s , q , response)
9. End For
10. End For
11. End For
12. End For
13. For each record (response, ground_truth) in results:
14. Compute ExactMatch(response, ground_truth)
15. Compute F1(response, ground_truth)
16. Compute SemanticSim(response, ground_truth) via lexical similarity
17. Compute STSim(response, ground_truth) via SentenceTransformer embeddings
18. End For
19. Aggregate metrics by (model, temperature, system instruction)
20. Perform two-way ANOVA for each dependent variable {F1, SemanticSim, STSim}
21. If ANOVA indicates significance ($p < 0.05$):
22. Conduct Tukey HSD post-hoc comparisons across factor levels
23. Visualize results using boxplots (system_ins) and lineplots (temperature)
24. Interpret the statistical outcomes to infer how t and s influence hallucination behavior

Return:

Final table and plots summarizing model-specific hallucination tendencies.

4. EXPERIMENTAL RESULTS AND DISCUSSION

The results obtained from the two-way analysis of variance and corresponding descriptive statistics reveal clear trends in how *system instruction* and *temperature* jointly influence model accuracy and semantic coherence across all metrics.

Effect of System Instruction

Across all metrics, the system instruction factor demonstrated a consistent influence on model performance. For the F1-score, the “*few-shot*” setting produced the highest mean value ($M = 0.189$), followed by “*say-IDK*” ($M = 0.181$) and “*zero-shot*” ($M = 0.172$). This suggests that providing a few examples improved the model’s ability to recall correct answers, likely by offering additional contextual grounding and reduced ambiguity. However, for semantic similarity, “*zero-shot*” achieved the highest mean ($M = 0.210$), marginally outperforming “*few-shot*” ($M = 0.206$). This implies that while few-shot prompts enhance precision, zero-shot prompts tend to yield semantically more coherent responses. A similar trend was observed in st-similarity, where “*zero-shot*” exhibited the highest mean ($M = 0.673$), compared to $M = 0.646$ for “*few-shot*” and $M = 0.645$ for “*say-IDK*”. This indicates that zero-shot prompting promotes more contextually consistent and embedding-level coherent responses. Collectively, these findings highlight a trade-off: Few-shot instructions enhance factual precision (F1), whereas zero-shot instructions strengthen semantic coherence (*st_similarity*). The interaction effects between temperature and system instruction are illustrated in Figure 3, 4 and 5, showing mean metric values across all factor combinations.

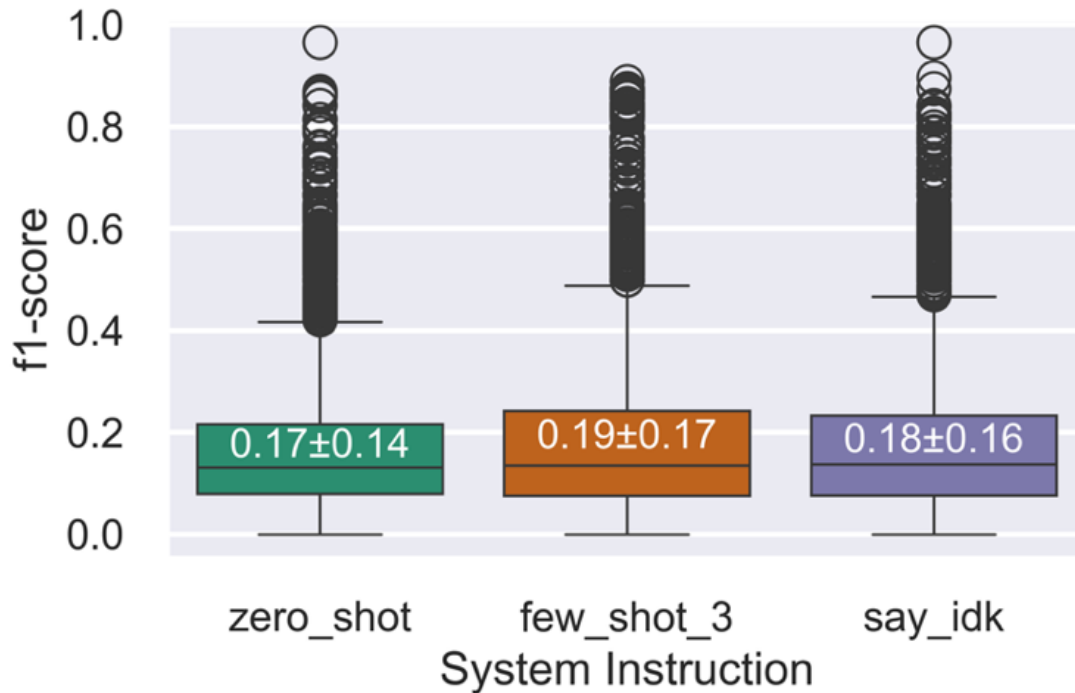


Figure 3. Distribution of F1-scores across system instruction conditions. Each box represents the interquartile range, the median is indicated by the horizontal line, and outliers are shown as points

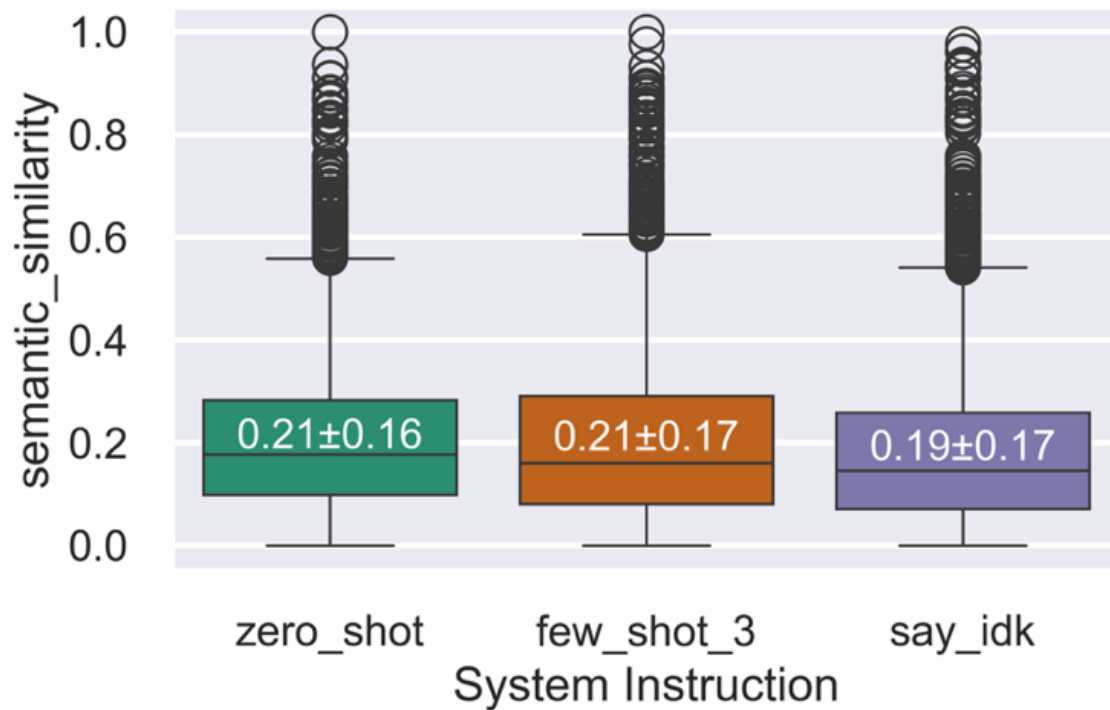


Figure 4. Distribution of TF-IDF semantic similarity across system instruction conditions. Boxplots summarize central tendency and variability for each category

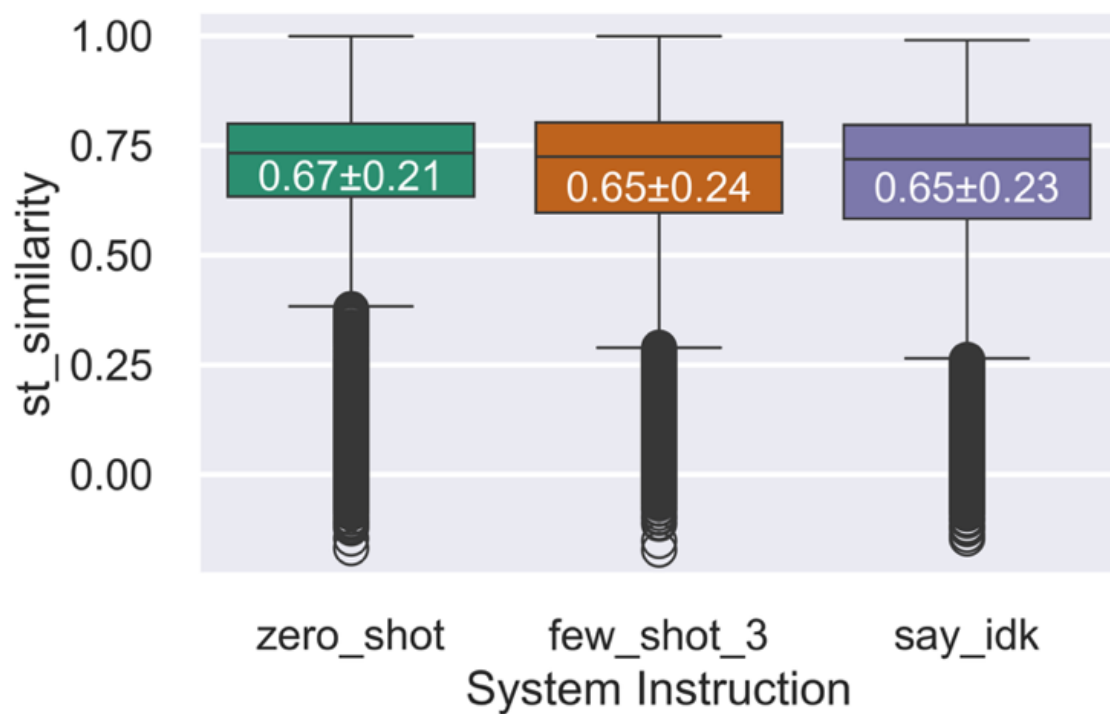


Figure 5. Distribution of ST-similarity scores across system instruction conditions. Boxplots highlight median, quartiles, and variability

Effect of Temperature

The restricted temperature range used in this study should be interpreted in light of its focus on factual reliability rather than generative diversity. While higher temperatures may yield richer or more creative outputs, they also amplify randomness in token selection and are therefore less representative of deployment scenarios where hallucination risk must be minimized. Consequently, the modest temperature effects observed here reflect behavior within a conservative decoding regime, complementing prior work that has examined more extreme temperature settings for creativity-oriented tasks. Temperature demonstrated a modest but systematic influence across metrics. As temperature increased from 0.0 to 0.7, all performance measures declined slightly. For instance, the mean F1-score dropped from 0.190 to 0.184 in the “*few-shot*” setting and 0.176 to 0.165 in “*zero-shot*.” Similarly, *st-similarity* decreased marginally across all configurations (e.g., 0.674 to 0.670 for “*zero-shot*”). This pattern aligns with theoretical expectations: higher temperatures increase sampling variability, leading to more diverse but less accurate generations and thus slightly elevated hallucination risk.

Interaction Between Temperature and System Instruction

The interaction between temperature and instruction remained small and statistically nonsignificant, which is visually confirmed by nearly parallel lines in the point plots. This suggests that the effect of system instruction generalizes across temperature levels. In other words, prompting strategy determines overall performance, whereas temperature fine-tunes the confidence–diversity balance. The interaction effects between temperature and system instruction are illustrated in Figure 6, 7 and 8, showing mean metric values across all factor combinations. From an applied perspective, these experimental results suggest the followings:

- “*zero-shot*” instruction is more robust for semantic stability and conceptual faithfulness.
- “*few-shot*” instruction improves literal correctness but may limit flexibility.
- Moderate or low temperature settings (≤ 0.2) are preferable when minimizing hallucination risk is a priority.

Two-Way ANOVA Results

To provide formal inferential support for the reported trends,

Table 3 summarizes the results of the two-way ANOVA conducted for each evaluation metric. This table reports degrees of freedom (DF), F-statistics, p-values, and effect sizes (η^2) for the main effects of temperature and system instruction, as well as their interaction. Across all metrics, system instruction exhibited a statistically significant main effect with the largest effect sizes, whereas temperature showed a smaller and metric-dependent influence. The interaction between temperature and system instruction was consistently non-significant, indicating that the effect of prompt strategy remained stable across temperature levels. Appendix Table A1 summarizes the Tukey HSD post-hoc comparisons for temperature and system instruction factors across all evaluation metrics; statistically significant differences are mainly associated with instruction

strategies rather than temperature levels, whereas non-significant contrasts, particularly under the *say-IDK* condition, reflect conservative abstention behavior rather than increased hallucination.

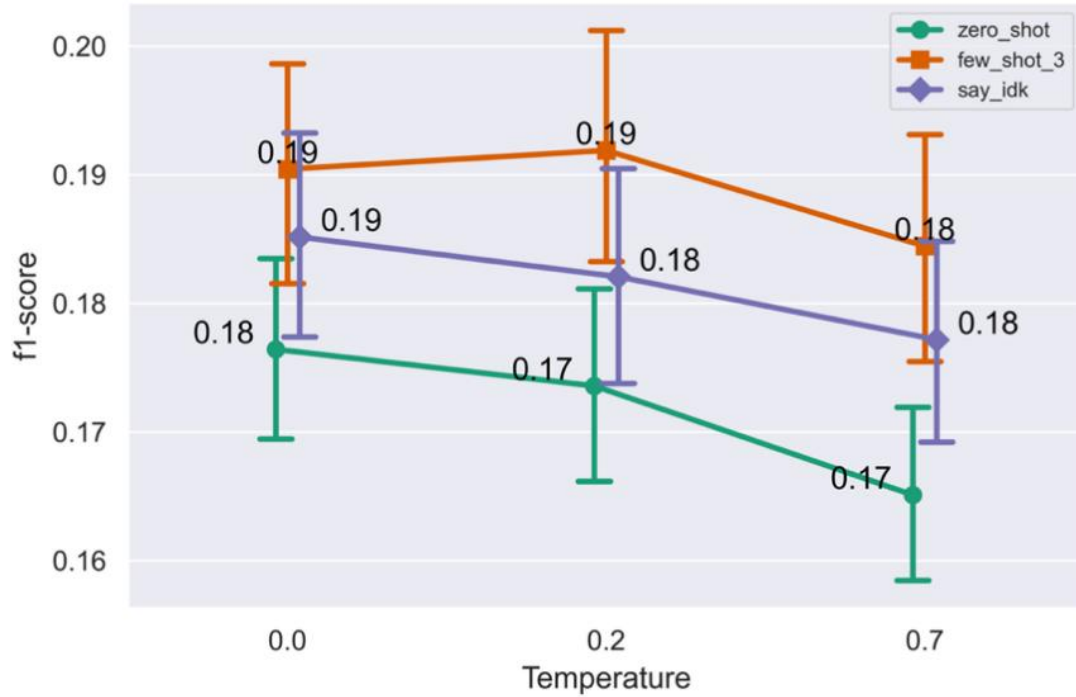


Figure 6. Interaction between temperature and system instruction for F1-scores. While points represent mean values, error bars represent standard error of the mean

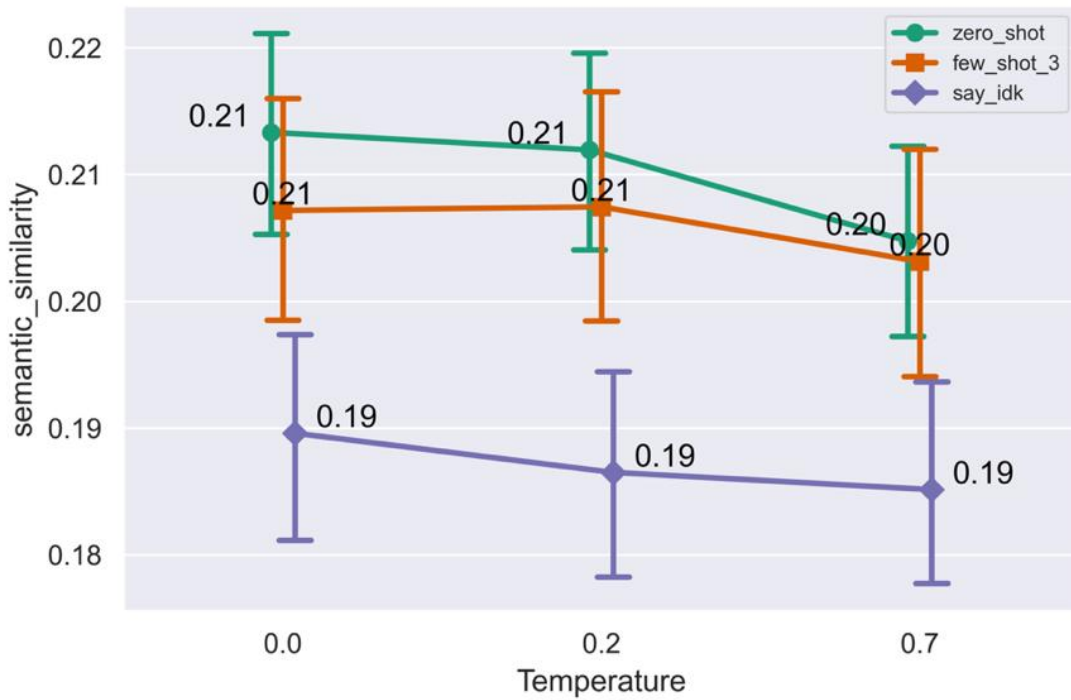


Figure 7. Interaction between temperature and system instruction for semantic similarity. While points represent mean values, error bars represent standard error of the mean

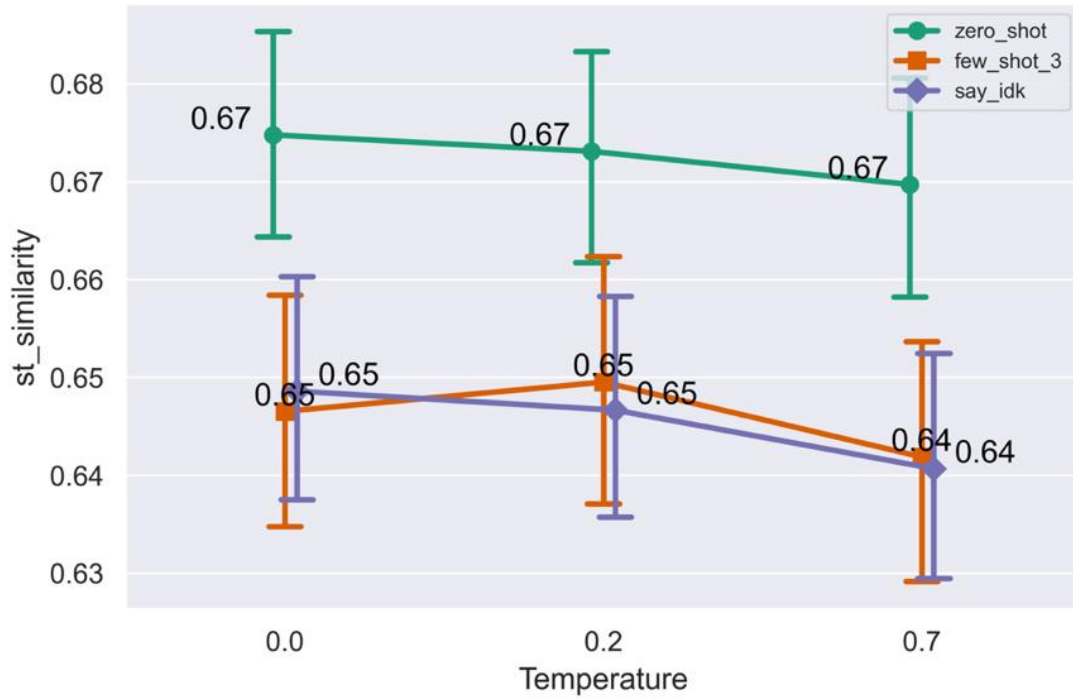


Figure 8. Interaction between temperature and system instruction for ST-similarity. Each point indicates the mean score for a given temperature and instruction combination

Table 3. Two-way ANOVA results for all evaluation metrics

Metric	Effect	DF	F	p-value	η^2
F1-Score	Temperature	2,13491	3.571	0.028	0.00053
	System Instruction	2,13491	13.154	< 0.001	0.00195
	Temperature $\times$ Instruction	4,13491	0.168	0.955	0.00005
Semantic Similarity	Temperature	2,13491	1.447	0.235	0.00021
	System Instruction	2,13491	24.635	< 0.001	0.00364
	Temperature $\times$ Instruction	4,13491	0.163	0.957	0.00005
ST Similarity	Temperature	2,13491	0.974	0.378	0.00014
	System Instruction	2,13491	21.080	< 0.001	0.00311
	Temperature $\times$ Instruction	4,13491	0.075	0.990	0.00002

Discussion

The results of this study provide a nuanced perspective on how decoding temperature and system instruction strategies influence hallucination-related behavior in lightweight open-source language models. Both factors exhibited measurable effects, but their relative magnitudes and practical implications differed substantially. Two-way ANOVA results showed that system instruction consistently produced statistically significant main effects across all evaluation metrics, with larger—though still modest—effect sizes ($\eta^2 \approx 0.002$ – 0.004) than those observed for temperature ($\eta^2 < 0.001$). These magnitudes are meaningful in light of the large sample size and the inherently stochastic nature of generative language models, where even small systematic shifts can accumulate into observable behavioral differences. The comparatively limited effect of temperature can be interpreted through the lens of probabilistic decoding mechanisms. Within the low-to-moderate temperature range examined in this study, token selection remains strongly dominated by high-probability continuations. As a result, increases in temperature primarily introduce minor lexical variability rather than substantial deviations in factual content or reasoning trajectories. This explains why temperature changes produced only marginal declines in factual and semantic alignment metrics and why no significant interaction effects with system instruction were observed. In contrast, higher temperature regimes—often explored in creativity-oriented studies—are more likely to amplify sampling randomness and hallucination risk, but such regimes were intentionally outside the scope of this reliability-focused analysis. Post-hoc analyses further clarified the role of instruction strategies. Few-shot prompting yielded significantly higher F1-scores than zero-shot prompting, suggesting that exemplar-based guidance helps align model outputs more closely with reference formulations. From a mechanistic perspective, this effect is consistent with in-context learning dynamics, where provided examples bias the model's conditional probability distribution toward answer patterns that resemble the demonstrations. Importantly, however, the data do not support claims that few-shot prompting constrains expressive or semantic flexibility. Rather, few-shot instructions appear to increase lexical conformity without conclusively limiting conceptual breadth, as evidenced by the absence of strong effects on embedding-based similarity metrics. Zero-shot prompting, by contrast, achieved higher semantic and embedding-based similarity scores, particularly relative to the *say-IDK* strategy. This pattern may reflect the reliance of instruction-tuned models on broadly internalized priors acquired during pretraining and alignment, allowing them to generate semantically coherent responses without explicit exemplars. However, these advantages were associated with small effect sizes and should therefore be interpreted as tendencies rather than robust dominance. The *say-IDK* strategy occupies a distinct position in this space: its lower scores on accuracy and similarity metrics do not indicate increased hallucination, but rather reflect deliberate abstention behavior that prioritizes epistemic caution over informativeness. Crucially, hallucination in this study was assessed indirectly through graded measures of factual and semantic alignment rather than as a binary phenomenon. Lower scores on strict metrics such as exact match or F1-score may arise from overt factual errors, but they may also result from correct yet non-literal responses or intentional non-response. Conversely, high embedding similarity captures conceptual proximity but does not guarantee factual correctness. By jointly analyzing multiple complementary metrics under a unified evaluation protocol, the study avoids conflating

stylistic variation, paraphrasing, and uncertainty-aware abstention with hallucination. The absence of statistically significant interaction effects further suggests that system instruction primarily shapes representational alignment, while temperature modulates token-level variability, with these factors operating largely independently. Taken together, these findings underscore that hallucination behavior in LLMs is governed by multiple, partially orthogonal mechanisms and that prompt strategies should be viewed as incremental control tools rather than definitive solutions.

5. CONCLUSION

This study systematically examined the effects of decoding temperature and system-level instruction strategies on the factual and semantic reliability of large language model outputs under a controlled experimental setting. Using lightweight, instruction-tuned models (*Gemma 2B*, *Mistral 7B Instruct*, and *Phi-3 Mini*) evaluated on the *TruthfulQA* benchmark, we observed that both sampling temperature and instruction framing produced measurable and statistically significant effects across factual (F1-scores), semantic, and sentence-embedding similarity metrics. These effects, while consistent, were associated with modest effect sizes, underscoring the inherently probabilistic and variable nature of generative language models. Within the examined temperature range, increases in temperature tended to reduce factual alignment while yielding only marginal changes in semantic similarity, reflecting a limited trade-off between lexical diversity and precision under reliability-oriented decoding regimes. System instruction strategies produced more stable and interpretable differences: few-shot prompting improved token-level accuracy through contextual grounding, zero-shot prompting favored semantic coherence, and the “*say-I-don’t-know*” strategy promoted conservative response behavior by discouraging overconfident assertions. Importantly, these outcomes should be interpreted as conditional patterns observed within the specific experimental configuration rather than as universal properties of language models or prompting techniques. The absence of significant interaction effects suggests that temperature and instruction strategies influenced distinct aspects of the generation process, sampling stochasticity and representational alignment, respectively, but this conclusion remains bounded by the scope of the models, tasks, and evaluation metrics employed. Consequently, the findings do not imply that hallucination mitigation can be fully achieved through any single control mechanism. Rather, they highlight how different design choices bias model behavior in incremental and context-dependent ways.

Future research is needed to assess the generalizability of these observations across larger and more diverse models, alternative benchmarks, and real-world deployment settings. Extensions to multimodal tasks, domain-specific applications, and adaptive decoding or instruction strategies would further clarify the extent to which these effects persist beyond the controlled conditions studied here. By maintaining a clear distinction between empirically supported findings and broader generalization, this work aims to contribute a cautious yet informative perspective on how decoding and instruction design shape hallucination-related behavior in contemporary language models.

APPENDIX

Table A1. Tukey HSD post-hoc comparison results

Metric	Factor	Group 1	Group 2	Mean Diff	p-adj	95% CI	Significant
F1-Score	System Instruction	few-shot	say-IDK	-0.0075	0.0689	[-0.0153, 0.0004]	No
		few-shot	zero-shot	-0.0172	< 0.001	[-0.0251, -0.0093]	Yes
		say-IDK	zero-shot	-0.0098	0.0105	[-0.0177, -0.0019]	Yes
	Temperature	0.0	0.2	-0.0015	0.8974	[-0.0094, 0.0064]	No
		0.0	0.7	-0.0084	0.0331	[-0.0163, -0.0005]	Yes
		0.2	0.7	-0.0069	0.0986	[-0.0148, 0.0010]	No
Semantic Similarity	System Instruction	few-shot	say-IDK	-0.0188	< 0.001	[-0.0270, -0.0107]	Yes
		few-shot	zero-shot	0.0041	0.4691	[-0.0041, 0.0123]	No
		say-IDK	zero-shot	0.0229	< 0.001	[0.0148, 0.0311]	Yes
	Temperature	0.0	0.2	-0.0014	0.9157	[-0.0096, 0.0068]	No
		0.0	0.7	-0.0057	0.2332	[-0.0139, 0.0025]	No
		0.2	0.7	-0.0043	0.4356	[-0.0125, 0.0039]	No
ST-Similarity	System Instruction	few-shot	say-IDK	-0.0007	0.9894	[-0.0119, 0.0105]	No
		few-shot	zero-shot	0.0265	< 0.001	[0.0153, 0.0378]	Yes
		say-IDK	zero-shot	0.0272	< 0.001	[0.0160, 0.0384]	Yes
	Temperature	0.0	0.2	-0.0002	0.9989	[-0.0114, 0.0110]	No
		0.0	0.7	-0.0059	0.4363	[-0.0171, 0.0053]	No
		0.2	0.7	-0.0057	0.4625	[-0.0169, 0.0056]	No

CONFLICT OF INTEREST

The author declares no conflict of interest.

REFERENCES

Abdin, M., Aneja, J., Awadalla, H., Awadallah, A., Awan, A. A., Bach, N., Bahree, A., Bakhtiari, A., Bao, J., Behl, H., Benhaim, A., Bilenko, M., Bjorck, J., Bubeck, S., Cai, M., Cai, Q., Chaudhary, V., Chen, D., Chen, D., ... Zhou, X. (2024). Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone. <https://doi.org/10.48550/arXiv.2404.14219>

- Chen, C., & Shu, K. (2023). *Can LLM-Generated Misinformation Be Detected?*. In: Proceedings of the 12th International Conference on Learning Representations (ICLR 2024). <https://doi.org/10.48550/arXiv.2309.13788>
- Coeckelbergh, M. (2025). LLMs, Truth, and Democracy: An Overview of Risks. *Science and Engineering Ethics*, 31(1), 4. <https://doi.org/10.1007/S11948-025-00529-0>
- Dahl, M., Magesh, V., Suzgun, M., & Ho, D. E. (2024). Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. *Journal of Legal Analysis*, 16(1), 64–93. <https://doi.org/10.1093/JLA/LAAE003>
- Datasets. (2025). Hugging Face. <https://huggingface.co/docs/datasets/en/index>
- DeepSeek. (2024). deepseek-llm. DeepSeek. https://ollama.com/library/deepseek-llm?utm_source=chatgpt.com
- Du, W., Yang, Y., & Welleck, S. (2025, July 13-19). *Optimizing Temperature for Language Models with Multi-Sample Inference*. In: A. Singh, M. Fazel, D. Hsu, S. Lacoste-Julien, F. Berkenkamp, T. Maharaj, K. Wagstaff, & J. Zhu (Eds.), Proceedings of the 42nd International Conference on Machine Learning (PMLR 267), (pp. 14648–14668), Vancouver, Canada. <https://proceedings.mlr.press/v267/du25f.html>
- Dziri, N., Milton, S., Yu, M., Zaiane, O., & Reddy, S. (2022). *On the Origin of Hallucinations in Conversational Models: Is it the Datasets or the Models?*. In: M. Carpuat, M.-C. de Marneffe, & I. V. M. Ruiz (Eds.), Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL 2022), (pp. 5271–5285), Seattle, United States. <https://doi.org/10.18653/v1/2022.naacl-main.387>
- Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., Casas, D. de las, Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L. R., Lachaux, M.-A., Stock, P., Scao, T. Le, Lavril, T., Wang, T., Lacroix, T., & Sayed, W. El. (2023). Mistral 7B. 2310.06825. <https://doi.org/10.48550/arXiv.2310.06825>
- Ladhak, F., Durmus, E., Suzgun, M., Zhang, T., Jurafsky, D., McKeown, K., & Hashimoto, T. (2023). *When Do Pre-Training Biases Propagate to Downstream Tasks? A Case Study in Text Summarization*. In: A. Vlachos, & I. Augenstein (Eds.), Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2023), (pp. 3206–3219), Dubrovnik, Croatia. <https://doi.org/10.18653/v1/2023.eacl-main.234>
- Lin, S., Hilton, J., & Evans, O. (2022). *TruthfulQA: Measuring How Models Mimic Human Falsehoods*. In: S. Muresan, P. Nakov, & A. Villavicencio (Eds.), Proceedings of the Annual Meeting of the 60th Association for Computational Linguistics, (vol. 1), (pp. 3214–3252), Dublin, Ireland. <https://doi.org/10.18653/v1/2022.acl-long.229>
- Ma, B., Wang, X., Hu, T., Haensch, A. C., Hedderich, M. A., Plank, B., & Kreuter, F. (2024). *The Potential and Challenges of Evaluating Attitudes, Opinions, and Values in Large Language Models*. In: Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), Findings of the Association for Computational Linguistics

- (EMNLP 2024), (pp. 8783–8805), Miami, Florida, USA. <https://doi.org/10.18653/V1/2024.FINDINGS-EMNLP.513>
- Ollama. (2025). Ollama. <https://ollama.com>
- Roustan, D., & Bastardot, F. (2025). The Clinicians' Guide to Large Language Models: A General Perspective With a Focus on Hallucinations. *Interactive Journal of Medical Research*, 14, e59823. <https://doi.org/10.2196/59823>
- Sahoo, P., Meharia, P., Ghosh, A., Saha, S., Jain, V., & Chadha, A. (2024). *A Comprehensive Survey of Hallucination in Large Language, Image, Video and Audio Foundation Models*. In: Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), Findings of the Association for Computational Linguistics (EMNLP 2024), (pp. 11709–11724), Miami, Florida, USA. <https://doi.org/10.18653/v1/2024.findings-emnlp.685>
- sentence-transformers/all-MiniLM-L6-v2. (2020). Hugging Face. <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>
- Shorinwa, O., Mei, Z., Lidard, J., Ren, A. Z., & Majumdar, A. (2025). A Survey on Uncertainty Quantification of Large Language Models: Taxonomy, Open Research Challenges, and Future Directions. *ACM Computing Surveys*, 58(3), 63. <https://doi.org/10.1145/3744238>
- Team, G., & Deepmind, G. (2024). Gemma: Open Models Based on Gemini Research and Technology.
- Wachter, S., Mittelstadt, B., & Russell, C. (2024). Do large language models have a legal duty to tell the truth? *Royal Society Open Science*, 11(8), 240197. <https://doi.org/10.1098/rsos.240197>
- Wei, J., Huang, D., Lu, Y., Zhou Quoc, D., & Le Google Deepmind, V. (2023). Simple synthetic data reduces sycophancy in large language models. <https://doi.org/10.48550/arXiv.2308.03958>
- Ye, H., Liu, T., Zhang, A., Hua, W., & Jia, W. (2024). *Cognitive Mirage: A Review of Hallucinations in Large Language Models*. In: Proceedings of the First International OpenKG Workshop: Large Knowledge-Enhanced Models Co-Located with The International Joint Conference on Artificial Intelligence (IJCAI 2024), (pp. 14–36). <https://github.com/hongbinye/Cognitive-Mirage-Hallucinations-in-LLMs>
- Zhang, M., Huang, M., Shi, R., Guo, L., Peng, C., Yan, P., Zhou, Y., & Qiu, X. (2024). *Calibrating the Confidence of Large Language Models by Eliciting Fidelity*. In: Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), 2024 Conference on Empirical Methods in Natural Language Processing, (pp. 2959–2979), Miami, Florida, USA. <https://doi.org/10.18653/V1/2024.EMNLP-MAIN.173>
- Zhang, Y., Li, J., & Li, W. (2023). *VIBE: Topic-Driven Temporal Adaptation for Twitter Classification*. In: H. Bouamor, J. Pino, & K. Bali (Eds.), 2023 Conference on Empirical Methods in Natural Language Processing, (pp. 3340–3354), Singapore. <https://doi.org/10.18653/v1/2023.emnlp-main.203>
- Zhang, Y., Li, S., Qian, C., Liu, J., Yu, P., Han, C., Fung, Y. R., McKeown, K., Zhai, C., Li, M., & Ji, H. (2025). *The Law of Knowledge Overshadowing: Towards Understanding, Predicting, and Preventing LLM Hallucination*. In: W. Che, J. Nabende, E. Shutova, & M. T. Pilehvar (Eds.), Findings of the Association for Computational Linguistics (ACL 2025), (pp. 23340–23358), Vienna, Austria. <https://doi.org/10.18653/v1/2025.findings-acl.1199>